

Bachelor of Technology — Electrical Engineering

Table 1: Electrical Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	52
	Discipline Electives	20
Electives	Free Electives	17
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		161

Table 2: B. Tech Electrical Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-210	Digital Systems Design	3-0-0-3	3
3	EE-210P	Digital Systems Design Lab	0-0-2-1	1
4	EE-260	Signals and Systems	2.5-0.5-0-3	3
5	EE-261	Electrical Systems Around Us	3-0-0-3	3
6	EE-261P	Electrical Systems Around Us Lab	0-0-3-2	2
7	EE-311	Device Electronics	3-0-0-3	3
8	EE-201	Electromechanics	2.5-0.5-0-3	3
9	EE-201P	Electromechanics Lab	0-0-2-1	1
10	EE-202	Electromagnetics & Wave Propagation	3-0-0-3	3
11	EE-211	Analog Circuit Design	2-0-2-3	3
12	EE-223P	Reverse Engineering	1-0-0-1	1
13	EE-304	Communication Systems	3-0-1-4	4
14	EE-231	Measurement and Instrumentation	2-0-2-3	3
15	EE-302	Control Systems	3-0-2-4	4
16	EE-303	Power System	3-1-0-4	4
17	EE-314	Digital Signal Processing	3-0-2-4	4
18	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
Total				52

Table 3: B. Tech Electrical Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		HS-342	German I	3-0-0-3	
10		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Electrical Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket (choose 1)	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9		IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	
10		IC-241	Material Science for Engineers	3-0-0-3	
11		IC-253	Data Structures and Algorithms	2-0-2-3	
12	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
13		HS-151	Introduction to English Literature	3-0-0-3	
14		HS-357	French Language and Culture	2-0-0-2	
15		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Electrical Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	EE-203	Network Theory	3-0-0-3	3
3	DC	EE-210	Digital Systems Design	3-0-0-3	3
4	DC	EE-210P	Digital Systems Design Lab	0-0-2-1	1
5	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
6	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
7	DC	EE-261P	Electrical Systems Around Us Lab	0-0-3-2	2
8	DC	EE-311	Device Electronics	3-0-0-3	3
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Electrical Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
3	DC	EE-201P	Electromechanics Lab	0-0-2-1	1
4	DC	EE-202	Electromagnetics & Wave Propagation	3-0-0-3	3
5	DC	EE-211	Analog Circuit Design	2-0-2-3	3
6	DC	EE-223P	Reverse Engineering	1-0-0-1	1
7	DC	EE-304	Communication Systems	3-0-1-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Electrical Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	EE-302	Control Systems	3-0-2-4	4
3	DC	EE-303	Power System	3-1-0-4	4
4	DC	EE-314	Digital Signal Processing	3-0-2-4	4
5	DC	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Electrical Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Electrical Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	EE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Electrical Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	EE-010	Internship		2
2	MTP	EE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 20 credits of **Discipline Electives** and 17 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Electrical Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BE-302	Bioelectric Systems Modeling	4
2	BE-304	Bioinformatics	4
3	CS-549	Performance Analysis of Computer Networks	3
4	CS-671	Deep Learning and Applications	4
5	EE-500	Network Control System	3
6	EE-501	Power System Operation and Control	3
7	EE-502	Analog System Design Laboratory	3
8	EE-503	Advance Communication Theory	3
9	EE-504	Switch Mode Power Conversion	3
10	EE-505	Materials for Semiconductor Industry	3
11	EE-506	Solar Photovoltaic Energy Systems	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	EE-507	Transmission Lines and Basic Microwave Engineering	4
13	EE-508	Fundamentals of Electric Drives	3
14	EE-508P	Practicum on fundamentals of Electrical Drives	2
15	EE-509	Linear Dynamical Systems	3
16	EE-511	Computer Vision	4
17	EE-512	CMOS Analog IC Design	4
18	EE-513	Special Electrical Machines	3
19	EE-514	Robust Control Systems	3
20	EE-515	Nonlinear Stability and Control	4
21	EE-516	Biomedical Systems	4
22	EE-517	Wireless Communications and Networking	3
23	EE-518	Information Theory	3
24	EE-519	CMOS Digital IC design Practicum	2
25	EE-519P	CMOS Digital IC Design Practicum	2
26	EE-520	Microelectronic Devices and Modelling	3
27	EE-522	Matrix Theory for Engineers	3
28	EE-523	Digital VLSI Architecture Design	4
29	EE-524	Digital MOS LSI Circuits	3
30	EE-526	Power Semiconductor Devices	3
31	EE-527	Analysis and Design of Power Electronic Converters	3
32	EE-527P	Practicum on Analysis and Design of Power Electronic Converters	2
33	EE-528	Modeling and Analysis of Electric Machines	3
34	EE-529	Embedded Systems	4
35	EE-530	Optimization theory	3
36	EE-531	Estimation and Detection Theory	3
37	EE-532	Supervised Research Exposure	3
38	EE-533	Access Network	4
39	EE-534	Probability and Random Processes	3
40	EE-535	Communication and Signal Processing Systems Design	3
41	EE-536	IoT Systems	3
42	EE-537	Power Quality Problems and Mitigation Techniques in Microgrids	3
43	EE-538	Data-driven Control	3
44	EE-540	Wide Band Gap Devices in Power Electronics Applications	3
45	EE-541	Sensors: Techniques, Algorithms, Applications for Signal	4
46	EE-542	Modelling, Simulation and Control of Hybrid Electric Vehicle	3
47	EE-543	Vision and Learning Based Control	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
48	EE-550	Optimal Control and Estimation	3
49	EE-551	Applied Photonics for Scientists and Engineers	3
50	EE-552	Power and Energy Systems	4
51	EE-553	Foundations of Intelligent Communication Systems-I	4
52	EE-554	Low Power VLSI Design	3
53	EE-555	Intelligent Control System	3
54	EE-556	Nuclear Reactor Control	1
55	EE-557	Adaptive Control	3
56	EE-560	Reconfigurable Computing	4
57	EE-570	Advanced Communication Theory	4
58	EE-574	Biomedical Signal and Image Analysis	3
59	EE-575	Applied Statistics for Data and Signals	4
60	EE-580	Network Systems: Modelling and Analysis	3
61	EE-581	Applied Statistics for Data and Signals	4
62	EE-582	Nonlinear Analysis and Control of Power Electronic Converters	3
63	EE-583	Smart Grids	3
64	EE-584	Power System Protection	3
65	EE-585	Theory, Modelling and Characterization of Photo-voltaic Devices	3
66	EE-593	Low Power VLSI Circuits	3
67	EE-597	Intellegent Control System	3
68	EE-601	Solid State Devices	3
69	EE-602	Control System Applications	3
70	EE-603	Renewable Energy and Smart Grid	3
71	EE-604	Practicum on Advanced Electric Drives	3
72	EE-604P	Practicum on Advanced Electric Drives	1
73	EE-605	Information Theory	3
74	EE-606	Introduction to High Voltage Engineering and Dielectric	3
75	EE-607	Optical Communication Systems	3
76	EE-608	Digital Image Processing	4
77	EE-609	Network Information Theory	3
78	EE-611	VLSI Technology	3
79	EE-611P	VLSI Fabrication Practicum	2
80	EE-612	OFDM For Optical Communications	3
81	EE-613	Wireless Communication	3
82	EE-614	Optical Communication Systems	3
83	EE-615	Nanoelectronics and Micronanofabrication	3
84	EE-616	Microwave engineering	2
85	EE-618	Industrial Process Control	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
86	EE-619	Mixed Signal VLSI Design	4
87	EE-620	Advanced Digital Signal Processing	3
88	EE-621	Radiating systems	4
89	EE-622	Microwave Integrated Circuits	3
90	EE-623P	Practicum on Digital Control of Power Electronics and Drives	3
91	EE-626P	Post Graduate Project - I	10
92	EE-630	HVDC Transmission and Flexible AC Transmission Systems	3
93	EE-641	Advance Wireless Technologies	4
94	EE-642	Research Study	3
95	EE-677	Analog Circuit Design	3
96	EE-678	Digital Circuit Design	3
97	EE-691	Advanced Topics in Dielectric Breakdown	1
98	EE-694	Photonics Computing	3
<i>Compulsory project components — counted as DE, not selectable</i>			
99	EE-498P	Major Technical Project - I (compulsory; counted as DE)	4
100	EE-499P	Major Technical Project - II (compulsory; counted as DE)	4